SDN BASED FIREWALL

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The controller receives packet_in events and installs flow rules dynamically. Matching is done based on IP and protocol fields. If traffic matches a blocked rule, a drop flow is installed.

1. ALLOWED TRAFFIC

```
vboxuser@Prajna:~$ sudo python3 topology.py
*** Configuring hosts
h1 h2 h3 h4
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Network started
*** H4 (10.0.0.4) is blocked by the firewall
*** Starting CLI:
mininet> h1 ping -c 3 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=4.12 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.530 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.056 ms

--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2038ms
rtt min/avg/max/mdev = 0.056/1.567/4.116/1.812 ms
mininet>
```

2. BLOCKED TRAFFIC

```
mininet> h1 ping -c 5 h4
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.

--- 10.0.0.4 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4096ms

mininet>
```

```
(ryu-env) vboxuser@Prajna:~/Desktop$ python -m ryu.cmd.manager firewall_controll
er.py
loading app firewall_controller.py
loading app ryu.controller.ofp_handler
instantiating app firewall_controller.py of SDNFirewall
instantiating app ryu.controller.ofp_handler of OFPHandler
[BLOCKED ICMP] 10.0.0.1 -> 10.0.0.4
```

The firewall was validated by testing both allowed and blocked scenarios. Allowed traffic was successfully forwarded, while blocked traffic resulted in packet drops and unreachable messages.

3. TCP BLOCKED

```
mininet> h4 python3 -m http.server 80 &
mininet> h1 wget http://10.0.0.4
--2026-04-15 00:39:19--  http://10.0.0.4/
Connecting to 10.0.0.4:80...
```

```
instantiating app ryu.controller.ofp_handler of
[BLOCKED TCP] 10.0.0.1 -> 10.0.0.4 port 80
```

4. FLOW TABLE

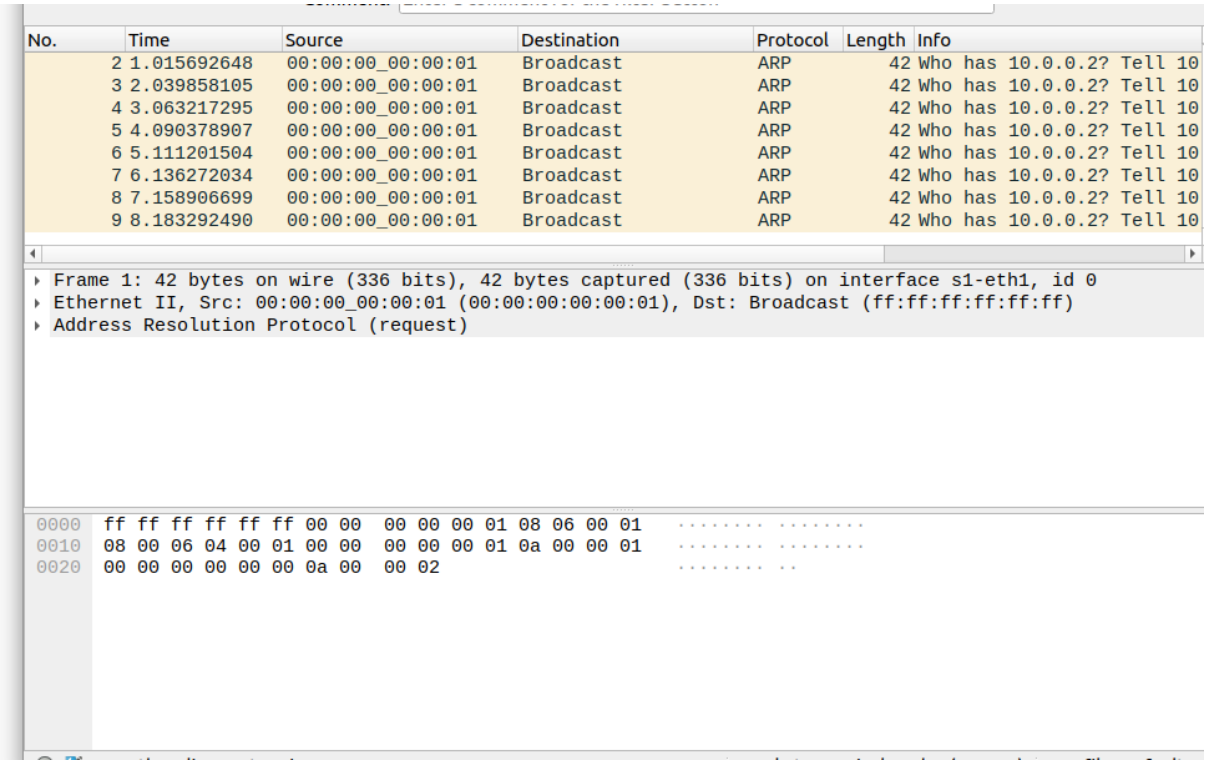
```
[BLOCKED TCP] 10.0.0.1 -> 10.0.0.4 port 80
^C(ryu-env) vboxuser@Prajna:~/Desktop$ sudo ovs-ofctl dump-flows s1
 cookie=0x0, duration=171.777s, table=0, n_packets=10, n_bytes=740, priority=30,tcp,nw
_dst=10.0.0.4,tp_dst=80 actions=drop
 cookie=0x0, duration=171.778s, table=0, n_packets=2, n_bytes=84, priority=10,in_port=
"s1-eth4",dl_dst=00:00:00:00:00:01 actions=output:"s1-eth1"
 cookie=0x0, duration=130.631s, table=0, n_packets=1, n_bytes=42, priority=10,in_port=
"s1-eth1",dl_dst=00:00:00:00:00:04 actions=output:"s1-eth4"
 cookie=0x0, duration=194.054s, table=0, n_packets=43, n_bytes=3298, priority=0 action
s=CONTROLLER:65535
(ryu-env) vboxuser@Prajna:~/Desktop$
```

5. PERFORMANCE TEST

```
mininet> h1 ping -c 5 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
From 10.0.0.1 icmp_seq=1 Destination Host Unreachable
From 10.0.0.1 icmp_seq=2 Destination Host Unreachable
From 10.0.0.1 icmp_seq=3 Destination Host Unreachable
From 10.0.0.1 icmp_seq=4 Destination Host Unreachable
From 10.0.0.1 icmp_seq=5 Destination Host Unreachable

--- 10.0.0.2 ping statistics ---
5 packets transmitted, 0 received, +5 errors, 100% packet loss, time 4094ms
pipe 4
mininet> h1 iperf -s &
mininet> h2 iperf -c h1
tcp connect failed: No route to host
------------------------------------------------------------
Client connecting to 10.0.0.1, TCP port 5001
TCP window size: -1.00 Byte (default)
------------------------------------------------------------
[  1] local 0.0.0.0 port 0 connected with 10.0.0.1 port 5001
mininet>
```

Proof Using Wireshark for allowed traffic:



No.	Time	Source	Destination	Protocol	Length	Info
2	1.015692648	00:00:00_00:00:01	Broadcast	ARP	42	Who has 10.0.0.2? Tell 10
3	2.039858105	00:00:00_00:00:01	Broadcast	ARP	42	Who has 10.0.0.2? Tell 10
4	3.063217295	00:00:00_00:00:01	Broadcast	ARP	42	Who has 10.0.0.2? Tell 10
5	4.090378907	00:00:00_00:00:01	Broadcast	ARP	42	Who has 10.0.0.2? Tell 10
6	5.111201504	00:00:00_00:00:01	Broadcast	ARP	42	Who has 10.0.0.2? Tell 10
7	6.136272034	00:00:00_00:00:01	Broadcast	ARP	42	Who has 10.0.0.2? Tell 10
8	7.158906699	00:00:00_00:00:01	Broadcast	ARP	42	Who has 10.0.0.2? Tell 10
9	8.183292490	00:00:00_00:00:01	Broadcast	ARP	42	Who has 10.0.0.2? Tell 10

```
▸ Frame 1: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface s1-eth1, id 0
▸ Ethernet II, Src: 00:00:00_00:00:01 (00:00:00:00:00:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▸ Address Resolution Protocol (request)

0000  ff ff ff ff ff ff 00 00  00 00 00 01 08 06 00 01   ........ ........
0010  08 00 06 04 00 01 00 00  00 00 00 01 0a 00 00 01   ........ ........
0020  00 00 00 00 00 00 0a 00  00 02                     ........ ..
```